

CBSE Class 12 Physics — Electrostatic Potential and Capacitance

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark Questions (Objective/VSA)

Q1. Which of the following is NOT a property of an equipotential surface? (a) They do not cross each other. (b) The rate of change of potential with distance on them is zero. (c) For a uniform electric field, they are concentric spheres. (d) They can be imaginary surfaces. [PYQ 2023] Very High

Q2. Equipotential surfaces: (a) are closer in regions of large electric fields compared to regions of lower electric fields. (b) will be more crowded near sharp edges of a conductor. (c) will never be equally spaced. (d) both (a) and (b) are correct. [PYQ 2022] High

Q3. The electric potential on the axis of a short electric dipole at a distance r from its centre is V . Then the potential at a point at the same distance on its equatorial line will be: (a) $2V$ (b) $-V$ (c) $V/2$ (d) Zero [PYQ 2022] Very High

Q4. If a unit positive charge is taken from one point to another over an equipotential surface: (a) work is done on the charge. (b) work is done by the charge. (c) work done is constant. (d) no work is done. [PYQ 2022] Very High

Q5. The electric potential V at any point (x, y, z) is given by $V = 3x^2$ where x is in metres and V in volts. The electric field at the point $(1 \text{ m}, 0.2 \text{ m})$ is: (a) 6 V/m along $(-x)$ -axis (b) 6 V/m along $(+x)$ -axis (c) 1.5 V/m along $(-x)$ -axis (d) 1.5 V/m along $(+x)$ -axis [PYQ 2021] Very High

Q6. An electric dipole of moment p is placed parallel to a uniform electric field E . The amount of work done in rotating the dipole by 90° is: (a) $2pE$ (b) pE (c) $pE/2$ (d) zero [PYQ 2021] High

Q7. Three charges $2q$, $-q$ and $-q$ lie at the vertices of an equilateral triangle. The value of the electric field E and electric potential V at the centroid of the triangle will be: (a) $E \neq 0$ and $V \neq 0$ (b) $E = 0$ and $V = 0$ (c) $E \neq 0$ and $V = 0$ (d) $E = 0$ and $V \neq 0$ [PYQ 2021] Medium

Q8. A solid spherical conductor has charge $+Q$ and radius R . It is surrounded by a solid spherical shell with charge $-Q$, inner radius $2R$, and outer radius $3R$. Which of the following statements is true? (a) The electric potential has a maximum magnitude at C and electric field has max magnitude at A . (b) Both the electric potential and electric field achieve a maximum magnitude at B . (c) The electric potential at A is zero and electric field has max magnitude at D . (d) The electric potential has a maximum magnitude at the center and electric field is maximum at the surface of the inner sphere. [PYQ 2021] High

Q9. Assertion (A): An electron has a higher potential energy when it is at a location associated with a more negative value of potential. **Reason (R):** Electrons move from a region of higher potential to a region of lower potential. [PYQ 2023] Very High

Q10. Assertion (A): Work done in moving a charge around a closed path in an electric field is always zero. **Reason (R):** Electrostatic force is a conservative force. [PYQ 2023] Very High

Q11. When a dielectric material is introduced between the plates of a charged and isolated parallel plate capacitor, the electric field between the plates: (a) decreases (b) remains constant (c) increases (d) first increases then decreases [PYQ 2022] Very High

Q12. Two parallel plate capacitors X and Y have the same area of plates and same separation. X has air and Y has a dielectric of constant 2. They are connected in series to a battery of 12 V. The ratio of electrostatic energy stored in X and Y is: (a) 4 : 1 (b) 1 : 4 (c) 2 : 1 (d) 1 : 2 [PYQ 2021] Medium

Q13. A capacitor's plates are charged by a battery with V volts. After charging, the battery is disconnected and a dielectric slab with dielectric constant K is inserted between its plates. The potential difference across the plates will become: (a) zero (b) $V/2$ (c) V/K (d) KV [PYQ 2021] Very High

Q14. A car battery is charged by a 12 V supply, and the energy stored in it is 7.20×10^5 J. The total charge passed through the battery is: (a) 6.0×10^4 C (b) 5.8×10^3 C (c) 8.64×10^6 C (d) 1.6×10^5 C [PYQ 2021] High

Q15. A variable capacitor is connected to a 200 V battery. If its capacitance is changed from $2\mu\text{F}$ to $X\mu\text{F}$, the decrease in energy of the capacitor is 2×10^{-2} J. The value of X is: (a) $1\mu\text{F}$ (b) $2\mu\text{F}$ (c) $3\mu\text{F}$ (d) $4\mu\text{F}$ [PYQ 2021] Medium

Q16. Name the physical quantity having SI unit $\text{N C}^{-1}\text{m}$. Is it a scalar or vector quantity? [PYQ 2020] High

Q17. Assertion (A): The capacitance of a parallel plate capacitor increases when a dielectric slab is introduced between its plates. **Reason (R):** The polarization of the dielectric reduces the net electric field, thereby reducing the potential difference for a given charge. [Practice] Very High

Q18. An air-filled parallel plate capacitor has a capacitance C . If the distance between the plates is doubled, its new capacitance becomes: (a) $2C$ (b) $C/2$ (c) $C/4$ (d) $4C$ [Practice] High

Q19. Does the electrostatic potential energy of a system of two positive charges increase or decrease when they are brought closer together? [Practice] Very High

Q20. Four charges $-q$, $-q$, $+q$, and $+q$ are placed at the corners of a square of side $2L$. What is the electric potential at the point exactly midway between the two $+q$ charges? [PYQ 2021] High

2-Mark Questions

Q21. Two charges 5×10^{-8} C and -3×10^{-8} C are located 16 cm apart. At what point on the line joining the two charges is the electric potential zero? (Take the potential at infinity to

be zero). [PYQ 2022] Very High

Q22. Write any two important characteristics of equipotential surfaces. [PYQ 2020] Very High

Q23. Draw equipotential surfaces due to an isolated point charge ($-q$) and depict the electric field lines. [PYQ 2020] High

Q24. Two closely spaced equipotential surfaces A and B with potentials V and $V + \delta V$ are kept δr distance apart. Deduce the relation between the electric field and the potential gradient between them. [PYQ 2014] High

Q25. Two point charges q and $-2q$ are kept d distance apart. Find the location of the point relative to charge q on the line joining them at which the electric potential due to this system of charges is zero. [PYQ 2014] High

Q26. Two charged conducting spheres of radii a and b are connected to each other by a wire. Find the ratio of the electric fields at their surfaces. [PYQ 2020] Very High

Q27. An electric field of 600 N/C is desired between two parallel capacitor plates kept 2 mm apart. What value of potential difference must be applied across the plates? [Practice] High

Q28. A 4000 pF air-gap capacitor is connected to a 30 V battery. If a piece of glass (dielectric constant $K = 9$) is placed between the plates to completely fill the gap, how much extra charge will flow from the battery? [Practice] Medium

Q29. A parallel plate capacitor of capacitance C is charged by a battery to voltage V . The battery is disconnected and an uncharged capacitor of capacitance $2C$ is connected in parallel across it. Find the ratio of the final charges on the two capacitors. [PYQ 2023] Very High

Q30. Explain why the electric field inside a solid conductor is zero under electrostatic conditions, and what this implies for the potential throughout the conductor's volume. [Practice] High

Q31. Can two equipotential surfaces intersect each other? Give reasons. [PYQ 2016] Very High

Q32. Find the expression for the potential energy of a system of two point charges q_1 and q_2 located at position vectors $\vec{r}_1$ and $\vec{r}_2$ respectively in an external uniform electric field $\vec{E}$. [PYQ 2020] Medium

3-Mark Questions

Q33. Derive an expression for the electric potential at any point (r, θ) along the axial line of an electric dipole. [PYQ 2017] Very High

Q34. Derive an expression for the capacitance of a parallel plate capacitor with air present between the two plates. [PYQ 2023] Very High

Q35. Derive an expression for the electrostatic energy stored in a charged parallel plate capacitor. [PYQ 2018] Very High

Q36. A parallel plate capacitor of capacitance C is charged to V volts. The battery is disconnected and the distance between the plates is doubled. A dielectric slab of dielectric

constant $K = 1.5$ is then introduced to fill the space. How will the (i) Electric field between the plates and (ii) Energy stored in the capacitor be affected? Justify. [PYQ 2019] High

Q37. Two capacitors $C_1 = 10\mu\text{F}$ and $C_2 = 15\mu\text{F}$ are connected in parallel, and the resulting combination is connected to a 12 V battery. Find: (a) The equivalent capacitance of the combination. (b) The potential difference across each capacitor. (c) The charge stored on each capacitor. [Practice] High

Q38. Draw equipotential surfaces for: (i) an electric dipole. (ii) two identical positive charges placed near each other. (iii) a uniform electric field in the z -direction. [PYQ 2022] Very High

Q39. Two identical capacitors of 12 pF each are connected in series across a 50 V battery. How much electrostatic energy is stored in the combination? If they were instead connected in parallel across the same battery, how much energy would be stored? [PYQ 2017] Medium

Q40. In a parallel plate capacitor with air between the plates, each plate has an area of $6 \times 10^{-3}\text{ m}^2$ and the separation is 3 mm . Calculate the capacitance. If connected to a 100 V supply, what is the charge on each plate? [PYQ 2022] Very High

Q41. A capacitor is connected across a source of potential difference V . The separation d between the plates is then steadily increased using an insulating handle. Plot a graph showing the variation of potential difference V versus plate separation d for this isolated capacitor. [Practice] High

Q42. Explain, using suitable diagrams, the difference in the behavior of a conductor and a dielectric in the presence of an external electric field. Define the term polarization of a dielectric. [PYQ 2015] High

5-Mark Questions (Long Answer/Case-Study)

Q43. (A) Derive an expression for the capacitance of a parallel plate capacitor when a dielectric slab of thickness t ($t < d$) is introduced between the plates. (B) A copper sphere of capacitance C is dropped into the ocean. Will the capacitance of the sphere increase, decrease, or remain the same? Justify. [PYQ 2025-26 Sample] Very High

Q44. A parallel plate capacitor of capacitance C is charged to a potential V by a battery. The battery remains connected to the circuit, and the plates are pulled apart to a larger distance. What changes will occur in the following quantities? Will they increase, decrease, or remain the same? Give an explanation in each case. (a) Capacitance (b) Charge (c) Potential difference (d) Electric field (e) Energy stored in the capacitor [PYQ 2025-26 Sample] Very High

Q45. (A) Two hollow spherical conductors of different sizes are charged positively. The smaller one is at 50 V and the larger one is at 100 V . If the smaller conductor is enclosed inside the larger conductor and they are connected by a conducting wire, in which direction will the charge flow? Explain with formulas. (B) The capacitance of an infinite parallel plate capacitor without a dielectric is C . It is half-filled with a dielectric medium of $K = 4$, first vertically (dividing plate area), and then horizontally (dividing plate separation). Prove that the

capacitance becomes $2.5C$ in the first case and $(4/5)C$ in the second case. [PYQ 2025-26 Sample] Very High

Q46. (A) Derive the expression for the potential energy of an electric dipole of dipole moment $\vec{p}$ in a uniform electric field $\vec{E}$. (B) An electric dipole of length 4 cm makes an angle of 60° with a uniform electric field and experiences a torque of $4\sqrt{3} \text{ Nm}$. Calculate the potential energy of the dipole in this orientation if it has charges of $\pm 8 \text{ nC}$. [Practice] High

Q47. (A) State the relationship between the electric field and the electric potential. (B) The electric potential at a point (x, y, z) is given by $V = -x^2y - xz^3 + 4$. Find the electric field vector $\vec{E}$ at that point. (C) A parallel plate capacitor C with a dielectric in between the plates is charged to a potential V by connecting it to a battery. The capacitor is then isolated. If the dielectric is suddenly withdrawn, will the energy stored in the capacitor increase or decrease? What causes this change in energy? [PYQ 2025-26 Sample] High

Q48. (A) Two identical small metal blocks each of mass m are held to each other by a metallic spring of force constant k . The block-spring system is placed on a frictionless surface and the spring has an unstretched length L_1 . A total charge Q is slowly given to the system, causing the spring to stretch to an equilibrium length L_2 . Assuming the charge distributes equally as point charges on the blocks, determine Q . (B) Derive the expression for the electric potential due to a point charge Q at a distance r from it. [PYQ 2025-26 Sample] High

Q49. (A) Define dielectric constant in terms of capacitance. (B) Two capacitors of unknown capacitances C_1 and C_2 are connected first in series and then in parallel across a 100 V battery. If the energy stored in the two combinations is 0.04 J and 0.25 J respectively, determine the values of C_1 and C_2 . (C) Calculate the charge on each capacitor in the parallel combination. [Practice] Medium

Q50. Case-Study: Dielectrics and Capacitance When a dielectric slab is inserted between the plates of a parallel plate capacitor, the external electric field polarizes the dielectric, creating an induced internal electric field that opposes the external field. The net electric field decreases, which reduces the potential difference across the plates (if isolated), thus increasing the capacitance by a factor of K . (a) What is the net electric field inside a dielectric of constant K if the original external field is E_0 ? (b) If a parallel plate capacitor has capacitance C_0 in air, what is its capacitance when completely filled with a conducting slab? (c) A $10\mu\text{F}$ capacitor is charged to 50 V and isolated from the battery. A dielectric slab of $K = 2$ is then fully inserted between the plates. Calculate the new potential difference across the plates and the final energy stored. [Practice] Very High

Answer Key

Q1: (c) For a uniform electric field, they are flat, parallel planes, not concentric spheres. **Q2:** (d) both (a) and (b) are correct. **Q3:** (d) Zero (Potential on the equatorial line of a dipole is always zero). **Q4:** (d) no work is done (Since $\Delta V = 0$, Work = $q\Delta V = 0$). **Q5:** (a) 6 V/m

along $(-x)$ -axis ($E_x = -dV/dx = -6x$. At $x = 1$, $E = -6$ V/m). **Q6:** (b) pE (Work $= \Delta U = -pE \cos(90^\circ) - (-pE \cos(0^\circ)) = pE$). **Q7:** (c) $E \neq 0$ and $V = 0$ (Potentials $+2V - V - V = 0$, but vectors of E don't cancel). **Q8:** (d) The electric potential has a maximum magnitude at the center. **Q9:** (c) Assertion is true but Reason is false. (Electrons, being negative, naturally move from lower to higher potential). **Q10:** (a) Both A and R are true, and R is the correct explanation. **Q11:** (a) decreases ($E = E_0/K$). **Q12:** (c) $2 : 1$ (In series, Q is same. $U = Q^2/2C$. Since $C_y = 2C_x$, $U_x/U_y = C_y/C_x = 2/1$). **Q13:** (c) V/K (Charge is conserved, C becomes KC , so $V' = Q/KC = V/K$). **Q14:** (a) 6.0×10^4 C ($E = QV \Rightarrow Q = E/V = 7.2 \times 10^5/12 = 60,000$ C). **Q15:** (a) $1\mu\text{F}$ (Initial $U = \frac{1}{2}(2\mu\text{F})(200)^2 = 0.04$ J. Decrease is 0.02 J, so final $U = 0.02$ J. $0.02 = \frac{1}{2}(X \times 10^{-6})(200)^2 \Rightarrow X = 1\mu\text{F}$). **Q16:** Electric potential. It is a scalar quantity. **Q17:** (a) Both A and R are true and R is the correct explanation. **Q18:** (b) $C/2$ ($C = \frac{\epsilon_0 A}{d}$, so $C \propto 1/d$). **Q19:** Increases (Since $U = k \frac{q_1 q_2}{r}$, as r decreases, U increases for like charges). **Q20:** $V = \frac{1}{4\pi\epsilon_0} \frac{2q}{L} \left(1 - \frac{1}{\sqrt{5}}\right)$. **Q21:** Let zero potential be at x from 5×10^{-8} C. $\frac{k(5 \times 10^{-8})}{x} = \frac{k(3 \times 10^{-8})}{16-x} \Rightarrow 5(16-x) = 3x \Rightarrow x = 10$ cm. **Q22:** (1) No work is done moving a charge on it. (2) Electric field is always perpendicular to it. **Q23:** Concentric spherical shells around $-q$. Field lines are straight lines pointing radially inward, perpendicular to the shells. **Q24:** $E = -\frac{\delta V}{\delta r}$. **Q25:** $\frac{kq}{x} + \frac{k(-2q)}{d-x} = 0 \Rightarrow \frac{1}{x} = \frac{2}{d-x} \Rightarrow d-x = 2x \Rightarrow x = d/3$ from charge q . **Q26:** $V_a = V_b \Rightarrow \frac{kQ_a}{a} = \frac{kQ_b}{b} \Rightarrow \frac{Q_a}{Q_b} = \frac{a}{b}$. Ratio of $E_a/E_b = \frac{Q_a/a^2}{Q_b/b^2} = \left(\frac{a}{b}\right)\left(\frac{b^2}{a^2}\right) = \frac{b}{a}$. **Q27:** $V = E \times d = 600 \times (2 \times 10^{-3}) = 1.2$ V. **Q28:** Initial $Q = CV = 4000 \text{ pF} \times 30 \text{ V} = 120 \text{ nC}$. Final $Q = KCV = 9 \times 120 = 1080 \text{ nC}$. Extra charge $= 1080 - 120 = 960 \text{ nC} = 0.96\mu\text{C}$. **Q29:** Voltage is common in parallel. $Q_1 = CV'$, $Q_2 = 2CV'$. Ratio is $1 : 2$. **Q30:** Charges in a conductor move freely until $E = 0$ internally. If $E = 0$, $dV/dr = 0$, meaning potential V is constant throughout the volume. **Q31:** No. If they intersected, there would be two different values of electric potential at the point of intersection, which is impossible. **Q32:** $U = q_1 V(\vec{r}_1) + q_2 V(\vec{r}_2) + \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|\vec{r}_1 - \vec{r}_2|}$. **Q33:** $V = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2 - a^2 \cos^2 \theta} \approx \frac{p}{4\pi\epsilon_0 r^2}$. **Q34:** State $E = \sigma/\epsilon_0$, $V = Ed = \frac{\sigma d}{\epsilon_0} = \frac{Qd}{A\epsilon_0}$. Thus, $C = Q/V = \frac{\epsilon_0 A}{d}$. **Q35:** Work done to add charge dq is $dW = Vdq = \frac{q}{C} dq$. Integrating from 0 to Q yields $U = \frac{Q^2}{2C} = \frac{1}{2} CV^2$. **Q36:** (i) $E = V'/d' = (V_0/1.5)/(2d) = E_0/3$. E field becomes $1/3$. (ii) Energy $U = Q^2/2C$. Q is constant, $C' = KC/2 = 0.75C$. $U' = Q^2/2(0.75C) = 1.33U_0$. **Q37:** (a) $C_{eq} = 10 + 15 = 25\mu\text{F}$. (b) $V = 12$ V across both. (c) $Q_1 = 10\mu\text{F} \times 12 \text{ V} = 120\mu\text{C}$; $Q_2 = 15\mu\text{F} \times 12 \text{ V} = 180\mu\text{C}$. **Q38:** (i) Dumbbell-shaped shells, (ii) Figure-8 shells merging outward, (iii) Parallel planes in xy plane. **Q39:** Series: $C_{eq} = 6 \text{ pF}$. $U = \frac{1}{2} \times 6 \times 10^{-12} \times 2500 = 7.5 \text{ nJ}$. Parallel: $C_{eq} = 24 \text{ pF}$. $U = \frac{1}{2} \times 24 \times 10^{-12} \times 2500 = 30 \text{ nJ}$. **Q40:** $C = \frac{8.85 \times 10^{-12} \times 6 \times 10^{-3}}{3 \times 10^{-3}} = 17.7 \text{ pF}$. $Q = CV = 17.7 \times 100 = 1770 \text{ pC} = 1.77 \text{ nC}$. **Q41:** Isolated means Q is constant. $C \propto 1/d$. $V = Q/C \propto d$. Graph is a straight line passing through the origin. **Q42:** Conductors cancel external E completely via free electron drift. Dielectrics only partially cancel it via molecular polarization. Polarization is the induced dipole moment per unit volume. **Q43:** (A) $V = E_0(d-t) + \frac{E_0}{K} t = \frac{Q}{A\epsilon_0} (d-t+t/K) \Rightarrow C = \frac{\epsilon_0 A}{d-t(1-1/K)}$. (B) It

will increase. The ocean acts as a dielectric surrounding the sphere, raising its capacitance ($C = 4\pi\epsilon_0 KR$). **Q44:** (a) C decreases ($C \propto 1/d$). (b) Q decreases ($Q = CV$, V is constant). (c) V remains the same (connected to battery). (d) E decreases ($E = V/d$). (e) U decreases ($U = \frac{1}{2}CV^2$, C drops). **Q45:** (A) Charge flows from smaller to larger shell. The inner shell potential is always higher ($V_i = V_{inner} + V_{outer}$ vs $V_o = V_{outer}$), so ΔV drives outward flow. (B) Vertical split (Parallel): $C' = C_1 + C_2 = \frac{\epsilon_0(A/2)}{d} + \frac{K\epsilon_0(A/2)}{d} = 0.5C + 2C = 2.5C$. Horizontal split (Series): $\frac{1}{C'} = \frac{d/2}{\epsilon_0 A} + \frac{d/2}{K\epsilon_0 A} \Rightarrow C' = \frac{2KC}{K+1} = \frac{8C}{5}$. **Q46:** (A) $dW = \tau d\theta = pE \sin \theta d\theta \Rightarrow U = -pE \cos \theta$. (B) $\tau = pE \sin 60^\circ \Rightarrow 4\sqrt{3} = pE \frac{\sqrt{3}}{2} \Rightarrow pE = 8 \text{ J}$. $U = -pE \cos 60^\circ = -8(0.5) = -4 \text{ J}$. **Q47:** (A) $\vec{E} = -\nabla V$. (B) $\vec{E} = -\hat{i}\frac{\partial V}{\partial x} - \hat{j}\frac{\partial V}{\partial y} - \hat{k}\frac{\partial V}{\partial z} = \hat{i}(2xy + z^3) + \hat{j}(x^2) + \hat{k}(3xz^2)$. (C) It will increase. Pulling the dielectric requires mechanical work against the attractive force of the plates, which converts into stored electrostatic energy ($U = Q^2/2C$, C drops so U rises). **Q48:** (A) Equilibrium: $F_{elec} = F_{spring} \Rightarrow \frac{1}{4\pi\epsilon_0} \frac{(Q/2)^2}{L_2^2} = k(L_2 - L_1)$. Solving gives $Q = 4L_2\sqrt{\pi\epsilon_0 k(L_2 - L_1)}$. (B) $dW = -E dr = -\frac{Q}{4\pi\epsilon_0 r^2} dr$, integrating from ∞ to r yields $V = \frac{Q}{4\pi\epsilon_0 r}$. **Q49:** (A) $K = C_{medium}/C_{vacuum}$. (B) $U_p = 0.25 \text{ J} \Rightarrow \frac{1}{2}(C_1 + C_2)(100)^2 = 0.25 \Rightarrow C_1 + C_2 = 50\mu\text{F}$. $U_s = 0.04 \text{ J} \Rightarrow C_s = 8\mu\text{F}$. $\frac{C_1 C_2}{50} = 8 \Rightarrow C_1 C_2 = 400$. Solving gives $10\mu\text{F}$ and $40\mu\text{F}$. (C) $Q_1 = 10\mu\text{F} \times 100 \text{ V} = 1000\mu\text{C}$; $Q_2 = 40\mu\text{F} \times 100 \text{ V} = 4000\mu\text{C}$. **Q50:** (a) $E = E_0/K$. (b) Infinity (since $K = \infty$ for a conductor, $E = 0$ inside, $\Delta V = 0$, $C \rightarrow \infty$). (c) Isolated so Q constant = $500\mu\text{C}$. C becomes $20\mu\text{F}$. New $V = 500\mu\text{C}/20\mu\text{F} = 25 \text{ V}$. Final $U = \frac{1}{2}C'V'^2 = \frac{1}{2}(20\mu\text{F})(25)^2 = 6250\mu\text{J} = 6.25 \text{ mJ}$.

CBSE Class 12 (2027) — Chapterwise PYQ Bank. Compiled from past board papers + board-pattern practice questions (clearly labelled).